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Will America Squander Its New Sputnik Moment?

By Walter G. Copan & Andrei Iancu

In 1957, the Soviet Union launched Sputnik 1 into orbit. It was a disheartening defeat for the United States, which was racing to launch its own satellite first. But losing the opening gambit in the Space Race spurred a resilient America into action. Washington doubled down on research and development, created legislative authorities to accelerate innovation, and ultimately put the first person on the moon in 1969.

For the two competing superpowers, space flight was initially a way to demonstrate military prowess. But winning the Space Race was about technological, economic, and political leadership as well. Inevitably, it came to signal capitalist democracy's superiority to communism.

Today, America faces a new Sputnik moment: It is being outdone in strategy, technology, and manufacturing by another authoritarian superpower—China. This time, though, it is not clear if the current setbacks will galvanize the United States into strategic action, or if Washington will cede superiority to Beijing. It depends on what policymakers do next—and whether the United States implements a muscular innovation strategy that works long term.

One big difference between the twentieth-century Sputnik moment and today is that, back then, the government was fully committed to technological advancement. Today, U.S. lawmakers are either distracted or, worse, considering steps that would hamstring American innovation.

Moreover, the Chinese Communist Party has learned from the Soviet collapse. In contrast to Moscow's nationalist insularity, China's strategic intent is to dominate research and education systems globally, harvest the world's innovations, manufacture them at the lowest cost, and control the technology standards that define markets and trade. Rewarded by shareholders, U.S. companies have willingly supported the Chinese strategy by outsourcing manufacturing and transferring their technologies abroad. With millennia of history and a demonstrated willingness to learn from it, China is now playing the long game to win. Simply put, China today offers a dramatic contrast to the Soviet Union of the Cold War era.

Another big difference is public awareness. In the early days of the Space Race, Americans followed every rocket launch and satellite with great, and even patriotic, interest. Today, distracted by two years of a global pandemic, many remain ignorant of just how far behind their country has fallen.

The United States Is Falling behind in Many Areas of Technology

Today's Sputnik moment is harder to see. It is not about just one critical technology, but many, and several are much harder to visualize than a rocket soaring into the sky.

Take, for example, silicon chips and other semiconductor components. Often tiny and hidden inside devices, they are not visible to most consumers, yet they power virtually every device, from dishwashers to cell phones, from pacemakers to automobiles. For almost a year now, the United States has faced a shortage of these essential components. Though America invented the semiconductor, it now manufactures only a fraction of the **globe's supply** and is largely dependent on foreign producers. **Taiwan**, the world's largest semiconductor manufacturer, remains a global flashpoint.

By pouring billions of dollars into **semiconductor manufacturing**, Beijing has managed to capture an ever-growing share of the market. Even more troubling, U.S. industry has embraced the outsourcing of semiconductor manufacturing, leaving Americans vulnerable to shortages when foreign suppliers do not deliver. Remarkably, the United States makes less than 13 percent of the world's supply (down from 37 percent in 1990), while Taiwan has over **63 percent** of the manufacturing market share. As part of its Made in China 2025 industrial policy, China has invested heavily with the goal of establishing a **position of leadership** in advanced semiconductors. As of 2019, **China's push** has resulted in chip sales reported from China to be higher than Taiwan's for the first time. Imagine the crisis if Taiwan were to fall to a military attack from mainland China.

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And semiconductors are not the United States' only vulnerability. The technologies that matter most in this tech race include 5G and 6G, artificial intelligence (AI), blockchain, quantum computing, and biopharmaceuticals. Unfortunately for the United States and its allies, as well as global prospects for democracy and free markets, America is currently losing ground in most of these technological areas. As a result, it is forced to rely on supply chains increasingly dominated by Chinese manufacturing in its centrally controlled economy, almost to the point of being hostage to it.

Consider 5G, the ultra-fast fifth generation of wireless connectivity. It will be the backbone of communications, the Internet of Things (IoT), and technologies such as virtual reality and autonomous vehicles. Extensively subsidized and supported by the Chinese government, tech giant **Huawei** is on a mission to dominate **5G networks** around the world and has signed over 90 international contracts to do so. As a common (but chilling) quip now goes, "it's Huawei or the highway."

China uses its influence and investments across both developed and developing nations for strategic advantage. For example, the communist nation has offered Latin American countries pandemic support in exchange for allowing Huawei to set up **networks**. Shortly after having requested vaccines from the Chinese government, **Brazil** allowed the company to bid on new 5G **network construction**.

This is cause for global concern because Chinese law requires its domestic companies to gather intelligence on behalf of the government. Huawei could use 5G networks to monitor critics, steal intellectual property (IP), collect information, or even disrupt internet access. Will they? Put another way, why would they not?

China is also a relentless surveillance state that is vying to become the world leader in AI after decades of U.S. dominance. Chinese firms lead the way in AI technologies, including robotics, facial recognition, and real-time translation. China has six times more patents than the United States in deep learning, the hottest field within AI, which allows machines to acquire information and make decisions.

China is likewise embracing cryptocurrencies, even as the United States holds back. In early 2021, Beijing unveiled a digital version of its currency, the yuan. Some Chinese policymakers hope the blockchain-enabled currency becomes so popular that it can one day replace the U.S. dollar as the world's reserve currency. Such tectonic change usually starts with small but noteworthy shifts: El Salvador recently became the world's first country to make Bitcoin legal tender.

Quantum technologies present great promise to revolutionize communications and computing with the power to break current cybersecurity encryption systems. This is not trivial: quantum computers will be many *millions* of times faster than the fastest supercomputers in use today. Although the United States had initial plans to demonstrate such a system, China was the first nation to demonstrate quantum communications in a satellite network.

The United States Is Still ahead in Biopharmaceuticals

Last century, the Soviet Union threatened U.S. leadership in space technology. In the twenty-first century, China threatens U.S. leadership in virtually all technologies of the future, with one notable exception. In biopharmaceuticals, the United States still has an appreciable advantage over China. Every year, the United States creates two-thirds of the world's new drugs, and it is responsible for more than 80 percent of all medicines in development. When Covid-19 began to ravage the globe, the United States led the way in cutting-edge vaccine development. China's Sinovac vaccines, by contrast, have proved notably unreliable.

U.S.-based companies created multiple life-saving vaccines in record time—including two based on state-of-the-art mRNA technology, which was largely developed by the United States and its European allies. Researchers anticipate that this same underlying mRNA technology will help treat many diseases, including HIV, multiple sclerosis, cystic fibrosis, Alzheimer's, and certain types of cancer.

China is working hard to catch up to the United States in drug development. Part of Beijing's *Made in China 2025* strategy includes a focus on becoming a major global competitor in biopharmaceuticals, seeking to erase America's lead in this area of technology as well. Its plan of action actually involves infiltrating the process by which U.S. researchers submit confidential research proposals to apply for grants from the National Institutes of Health and other organizations. This effort by China to illegally obtain American biomedical research plans has been investigated by the FBI and documented by major media outlets.

The United States has sent mixed signals on how it plans to compete. In his April 28, 2021, address to Congress, President Biden warned that "China and other countries are closing in fast. We have to develop and dominate the products and technologies of the future."

Yet, at the same time, his administration appears bent on diminishing U.S. competitiveness in biotechnology. Remarkably, the Biden administration has said it supports a petition before the World Trade Organization (WTO) to waive IP rights on Covid-19 vaccine technologies. It is clear that this action was intended to counter the substantial challenges in delivering Covid-19 vaccines and therapies worldwide.

However, not only would this be ineffective in solving the challenge of rapid global access to vaccines, but it would negatively impact the basis of future drug discovery while undermining the manufacturing network for these complex products around the world. This would effectively hand China and others the discoveries underlying these valuable innovations. To be clear: the United States cannot "develop and dominate the products and technologies of the future" if IP rights are not respected. Further, the humanitarian goal is to speed global access to vaccines and to end the pandemic and all its associated costs. An IP waiver would not actually speed up access to vaccines, given manufacturing limitations, and it could negatively impact response to the next crisis because of the way it would disincentivize innovation.

Giving up strategic advantage in one of the few areas where the United States still leads will have devastating consequences well beyond Covid-19 technologies. American entrepreneurship and innovation in all technology domains depend on the private sector. In turn, the private sector depends on IP rights to protect its investments.

The United States Must Protect Innovation

Since the founding of the nation, the **U.S. intellectual property system** has driven innovation and competitiveness. Patents and other IP protections give companies a limited time during which they have exclusive rights to their inventions—and a window of opportunity to recoup their investments. This chance to earn a return drives billions of dollars of annual private investment into research and development. Signaling that investors cannot rely on those protections—by, for example, supporting the WTO waiver petition—will deter private-sector investments and sacrifice America's advantage and ability to compete.

In another worrisome development, legislators have suggested **changing U.S. law** to reinterpret the decades-old Bayh-Dole Act, bedrock legislation for transforming federally funded research into commercial products. Their proposals would **strip IP rights** from companies that have invested millions to license and successfully develop university inventions.

The United States wishes to strengthen its competitive position globally. Yet the Biden administration has issued an **Executive Order (EO) on Promoting Competition** that, although well intentioned, includes provisions that would actually bring about the opposite effect.

The EO is intended to curb antitrust concerns across the economy. However, by weakening patent protections, the order would actually strengthen dominant companies' positions while reducing the ability of small and medium-sized companies to compete with them. Specifically, weakening patent protections in the pursuit of drug-price controls will reduce incentives to innovation and **undermine the competitive position** of America's biopharmaceutical industry, with collateral damage across all sectors important to the economy.

By weakening U.S. IP protections, including those **for standard-essential patents**, the EO directly supports **China's ambitions** to dominate, as laid out in *China Standards 2035*.

Recent court decisions have also invalidated virtually all U.S. patents on medical diagnostic technologies. **Similar decisions** have likewise put a cloud over computer and information-based intellectual properties and innovations. Congress has tried to initiate a legislative process to remedy this area of law, but all such efforts to date have stalled. It is no surprise that the United States is falling behind: without adequate IP protection on these and many other critical technologies, private investment and innovation dwindles. Because IP is one of the most important assets of entrepreneurial startup companies, if they do not have secure and enforceable IP rights, U.S. startups **will be increasingly unsuccessful in securing venture capital financing**.

These actions and inactions create, and perpetuate, a **U.S. innovation vacuum** that China and other nations are working vigorously to fill. China now files more patents than the United States, publishes more science and technical articles in peer-reviewed journals, graduates more science and engineering students, and works to lead more standards committees in the areas of technology that matter.

There are consequences to the decisions the United States makes as a country. Gutting intellectual property now would be like defunding NASA in the middle of the Space Race—and on top of it, handing over the blueprints for the Apollo spaceship and the Lunar Rover. China and other rivals fully recognize the importance of having a world-leading innovation economy. Russia even named its vaccine Sputnik—a clear sign that U.S. adversaries understand the economic, diplomatic, and even military power that stems from a nation's ability to quickly invent new medicines.

It is time for U.S. leaders to acknowledge this reality. The U.S. government must act to support, not undermine, American innovation. And it must do so immediately, for it will soon be too late.

There have been some encouraging developments. In June 2021, the Senate passed the United States Innovation and Competition Act, which includes over $100 billion in support for new technologies, including semiconductors, to be manufactured in the United States. Frustratingly, the House has not followed suit, and it is unlikely to do so in the near term.

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Substantially increasing the percentage of U.S. gross national product allocated to technological research and development would indeed be an important part of the response to China's threat. However, spending increases will be for naught unless the resulting innovations are protected. Specifically, the United States needs to strengthen its IP system and modernize technology transfer legislation for innovation and entrepreneurship to flourish here.

As with the Space Race, America must galvanize its immense resources and ingenuity to win this century's tech race. With the rapid development of Covid-19 vaccines, the United States achieved success on par with the moon landing. Yet, it still appears to be paralyzed in this present Sputnik moment. It is falling behind, with policymakers unsure of which way to go.

One path relinquishes technological leadership to China. The other gives democracy a chance to regain its global ascendance. The latter is far more preferable. ■

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